

Untitled model

Introduction to semantic modelling

A **concept scheme** (in SKOS: *ConceptScheme*) is a structured collection of concepts that carry a fixed, shared meaning within a particular domain. The goal is for everyone working with the model — people and systems alike — to use the same words for the same things. That sounds obvious, but in practice language causes a surprising amount of confusion: what one organisation calls a *measurement point*, another calls a *sensor*, a *location*, or an *observation station*.

A concept scheme solves this by recording, for each concept:

- a **preferred label** (prefLabel) it is known by,
- a **definition** that unambiguously describes what the concept does and does not cover,
- and optionally a **scope note** or **example** to clarify borderline cases.

Concepts rarely stand on their own. They are related **hierarchically** (a vehicle is broader than a truck) or **associatively** (a road user relates to a road segment). These relationships make it possible for queries and analyses to navigate between general and specific concepts, and to make connections between concepts explicit. Within SKOS we use `broader`, `narrower`, and `related` for that.

When a concept also exists elsewhere — in a national or European standard, for instance — we record that as a **mapping relation**. This prevents reinventing the wheel: we link to the source so interoperability with external systems remains possible.

This document describes the concept scheme as it is currently recorded. The following sections list the metadata, every concept, the relationships between them, and — where the model is connected to a data source — how it is linked to that source data.

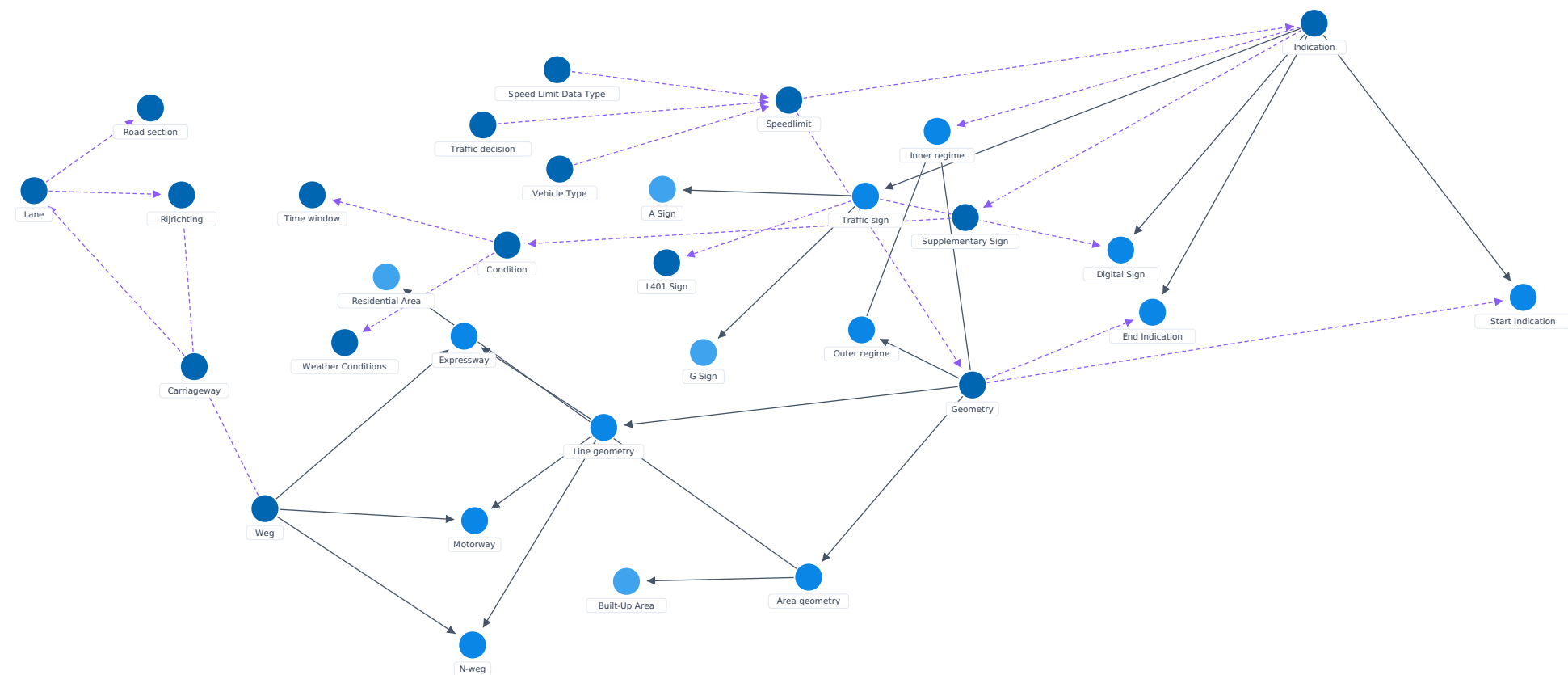
Naamloos model

Metadata

Field	Value
URI	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26
Version	0.0.1
Publication status	Draft
Management status	Active
Organization	Nationaal Data Portaal Wegverkeer
Created on	2026-07-17
Last modified on	2026-07-17
Last modified by	Arthur Keesen

Model overview

Visual overview of the concepts and their relations. Hierarchical relations are drawn with solid arrows; associative relations are drawn with dashed arrows.



→ broader / narrower (hiërarchisch / Associatief)
 - - - related (gerelateerd)

Bron: Ontologica — RTTI - Snelheidsbeperkingen (concept)

Concepts

Outer regime

Concept details

Field	Value
Definition	Nog bepalen
Example	A road user enters a built-up area (50 km/h). Subsequently, they enter a residential area (15 km/h). The active speed restriction is now 15 km/h. Upon leaving the residential area, 50 km/h applies again, because the built-up area as enclosing zone is still in effect.

Road section

Concept details

Field	Value
Definition	Nog bepalen

Residential Area

Concept details

Field	Value
Definition	Nog bepalen

Speedlimit

Concept details

Field	Value
Definition	Nog te bepalen

Expressway

Motorway

G Sign

Carriageway

Lane

Vehicle Type

Weather Conditions

Start Indication

End Indication

Indication

Digital Sign

Condition

Supplementary Sign

Built-Up Area

Inner regime

Concept details

Field	Value
Scope	The inner zone concept describes the cascading effect of nested zones. When entering an inner zone, the speed restriction of the enclosing zone is temporarily suspended. When exiting, the speed restriction of the enclosing zone becomes active again. This mechanism works recursively: multiple levels of nesting are possible.

Geometry

A Sign

Speed Limit Data Type

L401 Sign

Traffic sign

Concept details

Field	Value
Definition	The physical object — a panel (the 'plate'), normally mounted on a post or other support — bearing a traffic sign symbol, whose physical execution is legally prescribed.

Time window

Concept details

Field	Value
Definition	The days and/or hours indicated on a supplementary sign that imply a limitation of the temporal scope of application of the indication on the overlying traffic sign; in particular, for speed limits, the period during which the maximum speed indicated on the traffic sign applies.
Scope	The term 'time window' does not appear as such in the legislation; the legal basis lies in the limitation of the scope and is referred to as a 'time window' by Rijkswaterstaat and CROW. The concept has a dual realization: as a physical supplementary sign beneath a metal sign, or as a time condition within a digital/electronic sign. Not to be confused with 'window times' (loading and unloading times for urban logistics), which are incorporated into the Goods Delivery model.

<https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1772186214125>

<https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1772186485225>

Line geometry

Concept details

Field	Value
Definition	A one-dimensional geometric primitive representing a line: a continuous curve, measurable in length within a coordinate system, composed of one or more connected line segments in which the end point of each segment — except the last — coincides with the start point of the next. Line geometry is used to spatially represent linear features such as a road; designated GM_Curve in ISO 19107 and Curve/LineString in OGC Simple Features

Area geometry

Traffic decision

Concept details

Field	Value
Definition	

Field	Value
	Written decision by the competent authority deciding on the placement, modification, or removal of traffic signs and supplementary signs designated by Order in Council insofar as this creates or modifies a command or prohibition, or on measures on or along the road that lead to a restriction or expansion of the number of categories of road users.
Scope	Publication takes place pursuant to Article 26 of the BABW via the KOOP Traffic Decisions application in the Government Gazette (National Government) or the local publication journal (municipality, province, water board) since the entry into force of the Electronic Publications Act on 1 July 2021. Requirements for justification follow from Article 21 of the BABW; consultation with the Chief of Police follows from Article 24 of the BABW.

<https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1772187859423>

Relations per concept

Outgoing semantic relations only — these stay inside the model.

Outer regime

Semantic relations

Type	Target label	Target concept
broader	Geometry	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771237797860
narrower	Inner regime	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771237259923

Residential Area

Semantic relations

Type	Target label	Target concept
broader	Area geometry	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1772186704395

Speedlimit

Semantic relations

Type	Target label	Target concept
related	Geometry	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771237797860
related	Indication	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771236172013

Expressway

Semantic relations

Type	Target label	Target concept
broader	Line geometry	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1772186695677

Motorway

Semantic relations

Type	Target label	Target concept
broader	Line geometry	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1772186695677

Carriageway

Semantic relations

Type	Target label	Target concept
related	Rijrichting	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1772186214125
related	Lane	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771235638908

Lane

Semantic relations

Type	Target label	Target concept
related	Rijrichting	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1772186214125

Type	Target label	Target concept
related	Road section	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771235085917

Vehicle Type

Semantic relations

Type	Target label	Target concept
related	Speedlimit	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771235237134

Start Indication

Semantic relations

Type	Target label	Target concept
broader	Indication	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771236172013

End Indication

Semantic relations

Type	Target label	Target concept
broader	Indication	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771236172013

Indication

Semantic relations

Type	Target label	Target concept
narrower	Traffic sign	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1772185159176
narrower	Digital Sign	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771236355335
related	Inner regime	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771237259923
related	Supplementary Sign	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771236462476

Condition

Semantic relations

Type	Target label	Target concept
related	Weather Conditions	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771236028582
related	Time window	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1772186160984

Supplementary Sign

Semantic relations

Type	Target label	Target concept
related	Condition	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771236446358

Built-Up Area

Semantic relations

Type	Target label	Target concept
broader	Area geometry	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1772186704395

Inner regime

Semantic relations

Type	Target label	Target concept
broader	Geometry	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771237797860

Geometry

Semantic relations

Type	Target label	Target concept
related	End Indication	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771236142216
related	Start Indication	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771236090614

Speed Limit Data Type

Semantic relations

Type	Target label	Target concept
related	Speedlimit	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771235237134

Traffic sign

Semantic relations

Type	Target label	Target concept
narrower	A Sign	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771238610163

Type	Target label	Target concept
narrower	G Sign	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771235385932
related	Digital Sign	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771236355335
related	L401 Sign	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771857141294

<https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1772186485225>

Semantic relations

Type	Target label	Target concept
narrower	N-weg	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1772187859423
narrower	Expressway	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771235313383
narrower	Motorway	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771235331951
related	Carriageway	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771235620358

Line geometry

Semantic relations

Type	Target label	Target concept
broader	Geometry	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771237797860

Area geometry

Semantic relations

Type	Target label	Target concept
broader	Geometry	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771237797860

Traffic decision

Semantic relations

Type	Target label	Target concept
related	Speedlimit	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1771235237134

<https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1772187859423>

Semantic relations

Type	Target label	Target concept
broader	Line geometry	https://xcalibur1978.github.io/temp/id/fc47556b-385f-45ac-94e4-828536c3cb26/concept-1772186695677